PRN – 260245920034

1. *Count total no. of employees.*

```
mysql> select count(*) as TotalEmployees from emp;
+----------------+
| TotalEmployees |
+----------------+
|             30 |
+----------------+
1 row in set (0.00 sec)
```

2. *Determine the maximum and minimum salary.*

```
mysql> select max(sal) as MaximumSalary, min(sal) as MinimumSalary from emp;
+---------------+---------------+
| MaximumSalary | MinimumSalary |
+---------------+---------------+
|          5000 |           800 |
+---------------+---------------+
1 row in set (0.00 sec)
```

3. *Display the count of employees having salary greater than 3000.*

```
mysql> select count(*) as NumEmployees
    -> from emp
    -> where sal > 3000;
+--------------+
| NumEmployees |
+--------------+
|            4 |
+--------------+
1 row in set (0.00 sec)
```

4. *Print department wise count of employees.*

```
mysql> select d.deptno, d.dname, count(e.empno) as NumEmployees
            from dept d
            left join emp e  on d.deptno = e.deptno
            group by d.deptno, d.dname;
+--------+------------+--------------+
| deptno | dname      | NumEmployees |
+--------+------------+--------------+
|     10 | ACCOUNTING |            5 |
|     20 | RESEARCH   |            8 |
|     30 | SALES      |           12 |
|     40 | OPERATIONS |            0 |
|     50 | HR         |            3 |
|     60 | IT         |            2 |
+--------+------------+--------------+
6 rows in set (0.00 sec)
```

5. *Display employee details who earn maximum and minimum salary.*

-- Employees with Maximum Salary

mysql> select * from emp where sal = (select max(sal) from emp);

-- Employees with Minimum Salary

mysql> select * from emp where sal = (select min(sal) from emp);

```
mysql> select *
    -> from emp
    -> where sal = (select max(sal) from emp);
+-------+-------+--------+----------------+------+------------+------+------+--------+---------+------------+-----------+------------+----------+
| EMPNO | ENAME | GENDER | JOB            | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME  | PWD       | PHONE      | isActive |
+-------+-------+--------+----------------+------+------------+------+------+--------+---------+------------+-----------+------------+----------+
|  7839 | KING  | M      | PRESIDENT      | NULL | 1981-11-17 | 5000 | NULL |     10 |       1 | KING       | r50mpm    | 7132300081 |        1 |
|  7840 | FRED  | M      | vice PRESIDENT | 7839 | 1983-08-09 | 5000 | NULL |     20 |       1 | SAND STONE | a12recmpm | NULL       |        1 |
+-------+-------+--------+----------------+------+------------+------+------+--------+---------+------------+-----------+------------+----------+
2 rows in set (0.00 sec)

mysql> select *
    -> from emp
    -> where sal = (select min(sal) from emp);
+-------+-------+--------+-------+------+------------+------+------+--------+---------+-----------+-----------+------------+----------+
| EMPNO | ENAME | GENDER | JOB   | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME | PWD       | PHONE      | isActive |
+-------+-------+--------+-------+------+------------+------+------+--------+---------+-----------+-----------+------------+----------+
|  7369 | SMITH | M      | CLERK | 7902 | 1980-12-17 |  800 | NULL |     20 |       2 | TRITONE   | a12recmpm | 7032300023 |        0 |
+-------+-------+--------+-------+------+------------+------+------+--------+---------+-----------+-----------+------------+----------+
1 row in set (0.00 sec)
```

6. *Print jobwise total salary.*

mysql> select job, sum(sal) as TotalSalary from emp group by job;

```
+---------------------+-------------+
| job                 | TotalSalary |
+---------------------+-------------+
| Compliance officers |        5250 |
| Public Relation     |        3500 |
| CLERK               |       11000 |
| SALESMAN            |        9525 |
| MANAGER             |       13675 |
| ANALYST             |        7750 |
| PRESIDENT           |        5000 |
| vice PRESIDENT      |        6000 |
| Network             |        2000 |
+---------------------+-------------+
9 rows in set (0.00 sec)
```

7. *Print department wise maximum salary.*

mysql> select deptno, max(sal) as MaxSalary from emp group by deptno;

```
+--------+-----------+
| deptno | MaxSalary |
+--------+-----------+
|     10 |      5000 |
|     20 |      5000 |
|     30 |      4150 |
|     50 |      1750 |
|     60 |      2000 |
+--------+-----------+
5 rows in set (0.00 sec)
```

8. *Print jobwise average salary.*

```
mysql> select job, avg(sal) as AvgSalary from emp group by job;
+---------------------+-----------+
| job                 | AvgSalary |
+---------------------+-----------+
| Compliance officers | 1750.0000 |
| Public Relation     | 1750.0000 |
| CLERK               | 1571.4286 |
| SALESMAN            | 1587.5000 |
| MANAGER             | 2735.0000 |
| ANALYST             | 2583.3333 |
| PRESIDENT           | 5000.0000 |
| vice PRESIDENT      | 3000.0000 |
| Network             | 2000.0000 |
+---------------------+-----------+
9 rows in set (0.00 sec)
```

9. *Print count of employee working in department 20.*

```
mysql> select count(*) as NumEmployees from emp where deptno = 20;
+---------------------+
| NumEmployees |
+--------------------+
|        8            |
+-------------------+
1 row in set (0.02 sec)
```

10. *Print count of employee working in department 10 having job as MANAGER..*

```
mysql> select count(*) as NumEmployees from emp where deptno = 10  and job = 'MANAGER';
+---------------------+
| NumEmployees |
+--------------------+
|        1            |
+-------------------+
1 row in set (0.04 sec)
```

11. *Print count of employee working in department 20 having comm as null.*

```
mysql> select count(*) as NumEmployees from emp where deptno = 20  and comm is null;
+---------------------+
| NumEmployees |
+--------------------+
|        8            |
+-------------------+
1 row in set (0.03 sec)
```

12. *Print names of employees working in ACCOUNTS department having maximum salary.*
    mysql> select ename, sal from emp where deptno = (select deptno from dept where dname = 'ACCOUNTING') and sal = (select max(sal) from emp where deptno = (select deptno from dept where dname = 'ACCOUNTING'));

```
+-------+------+
| ename | sal  |
+-------+------+
| KING  | 5000 |
+-------+------+
1 row in set (0.00 sec)
```

13. *Print employee details having salary less than average salary of MANAGER.*
    mysql> select * from emp where sal < (select avg(sal) from emp where job = 'MANAGER');

```
mysql> select * from emp where sal < (select avg(sal) from emp where job = 'MANAGER');
+-------+----------+--------+---------------------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
| EMPNO | ENAME    | GENDER | JOB                 | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME    | PWD        | PHONE      | isActive |
+-------+----------+--------+---------------------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
|  6001 | GITA     | F      | Compliance officers | 7920 | 1981-07-19 | 1750 | NULL |     30 |       1 | GITA         | sales@2017 | NULL       |        1 |
|  6129 | VRUSHALI | F      | Compliance officers | 7920 | 1981-07-19 | 1750 | NULL |     50 |       1 | VRUSHALI     | PUN@8877   | NULL       |        1 |
|  6473 | SHARMIN  | F      | Public Relation     | 7920 | 1981-07-19 | 1750 | NULL |     20 |       1 | SHARMIN      | a12recmpm  | NULL       |        1 |
|  6781 | BANDISH  | F      | Public Relation     | 7920 | 1981-07-19 | 1750 | NULL |     20 |       1 | BANDISH      | a12recmpm  | NULL       |        1 |
|  7369 | SMITH    | M      | CLERK               | 7902 | 1980-12-17 |  800 | NULL |     20 |       2 | TRITONE      | a12recmpm  | 7032300023 |        0 |
|  7421 | THOMAS   | M      | CLERK               | 7920 | 1981-07-19 | 1750 | NULL |     10 |       1 | THOMAS       | r50mpm     | 7032300011 |        0 |
|  7499 | ALLEN    | M      | SALESMAN            | 7698 | 1981-02-20 | 1600 |  300 |     30 |       4 | ALWAYS TESTE | sales@2017 | 7032300096 |        1 |
|  7521 | WARD     | M      | SALESMAN            | 7698 | 1981-02-22 | 1250 |  500 |     30 |       1 | WARD         | sales@2017 | 7132300034 |        1 |
|  7654 | MARTIN   | M      | SALESMAN            | 7698 | 1981-09-28 | 1250 | 1400 |     30 |       6 | LIFE RACER   | sales@2017 | 7132300050 |        1 |
|  7782 | CLARK    | M      | MANAGER             | 7839 | 1981-06-09 | 2450 | NULL |     10 |       3 | CLARK        | r50mpm     | 7032300001 |        1 |
|  7844 | TURNER   | M      | SALESMAN            | 7698 | 1981-09-08 | 1500 |    0 |     30 |       5 | SAND DUST    | sales@2017 | NULL       |        1 |
|  7876 | ADAMS    | M      | CLERK               | 7788 | 1983-01-12 | 1100 | NULL |     20 |       1 | CRAZY ANYONE | a12recmpm  | 7132300086 |        1 |
|  7888 | Ruhan    | M      | Network             | 7888 | 1983-08-09 | 2000 | NULL |     60 |       1 | Ruhan        | BRD@1122   | NULL       |        1 |
|  7889 | Neel     | M      | vice PRESIDENT      | 7839 | 1983-08-09 | 1000 | NULL |     60 |       0 | Neel         | BRD@1122   | NULL       |        1 |
|  7900 | JAMES    | M      | CLERK               | 7698 | 1981-12-03 |  950 | NULL |     30 |       2 | CRAZY LEADER | sales@2017 | 7132300042 |        1 |
|  7920 | GRASS    | M      | SALESMAN            | 7919 | 1980-02-14 | 2575 | 2700 |     30 |       5 | GRASS        | sales@2017 | NULL       |        1 |
|  7934 | MILLER   | M      | CLERK               | 7782 | 1982-01-23 | 1300 | 1300 |     10 |       2 | QUARTERNOTE  | r50mpm     | 7132300055 |        0 |
|  7945 | AARUSH   | M      | SALESMAN            | 7902 | 1980-02-14 | 1350 | 2700 |     30 |    NULL | AARUSH       | ABCDEF     | NULL       |        0 |
|  7949 | ALEX     | M      | MANAGER             | 7698 | 1982-01-24 | 1250 |  500 |     30 |       5 | QUARTERREST  | sales@2017 | 7156567183 |        1 |
|  8433 | NATASHA  | F      | CLERK               | 7920 | 1981-07-19 | 1750 | NULL |     30 |       1 | NATASHA      | sales@2017 | NULL       |        1 |
|  9400 | SANGITA  | F      | Compliance officers | 7920 | 1981-07-19 | 1750 | NULL |     50 |       1 | SANGITA      | NULL       | NULL       |        1 |
|  9473 | SUPRIYA  | F      | ANALYST             | 7920 | 1981-07-19 | 1750 | NULL |     50 |       1 | SUPRIYA      | XYZ@1234   | NULL       |        1 |
+-------+----------+--------+---------------------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
22 rows in set (0.04 sec)
```

14. *Give SQL statement to find the average annual salary per job in each detp.*
    mysql> select deptno, job, avg(sal * 12) as AvgAnnualSalary from emp group by deptno, job;

```
+--------+---------------------+-----------------+
| deptno | job                 | AvgAnnualSalary |
+--------+---------------------+-----------------+
|     30 | Compliance officers |     21000.0000 |
|     50 | Compliance officers |     21000.0000 |
|     20 | Public Relation     |     21000.0000 |
|     20 | CLERK               |     11400.0000 |
|     10 | CLERK               |     25600.0000 |
|     30 | SALESMAN            |     19050.0000 |
|     20 | MANAGER             |     35700.0000 |
|     30 | MANAGER             |     33000.0000 |
|     10 | MANAGER             |     29400.0000 |
|     20 | ANALYST             |     36000.0000 |
|     10 | PRESIDENT           |     60000.0000 |
|     20 | vice PRESIDENT      |     60000.0000 |
```

```
|    60 | Network          |     24000.0000 |
|    60 | vice PRESIDENT     |     12000.0000 |
|    30 | CLERK            |     16200.0000 |
|    50 | ANALYST          |     21000.0000 |
+--------+--------------------+----------------+
```

15. *Count the number of people in the dept 30 who receive a salary and the no.of people who receive comm.*
    mysql> select count(sal) as NumWithSalary, count(comm) as NumWithComm from emp where deptno = 30;

```
+---------------+---------------------------+
| NumWithSalary | NumWithComm |
+---------------+---------------------------+
|          12 |         7                          |
+---------------+---------------------------+
1 row in set (0.00 sec)
```

16. *Calculate the avg, min and max salary of those groups of employees having the job as CLERK or MANAGER.*
    mysql> select job, avg(sal) as AvgSalary, min(sal) as MinSalary, max(sal) as MaxSalary from emp where job in ('CLERK', 'MANAGER') group by job;

```
+---------+-----------+-----------+-----------+
| job     | AvgSalary | MinSalary | MaxSalary |
+---------+-----------+-----------+-----------+
| CLERK   | 1571.4286 |      800 |      3350 |
| MANAGER | 2735.0000 |      1250 |      4150 |
+---------+-----------+-----------+-----------+
2 rows in set (0.00 sec)
```

17. *Display the deptno of departments which have more than one CLERK.*
    mysql> select deptno from emp where job = 'CLERK' group by deptno having count(*) > 1;

```
+--------+
| deptno |
+--------+
|    10 |
|    20 |
|    30 |
+--------+
3 rows in set (0.04 sec)
```

18. *List names and hiredates of employees who were hired in the month of December*
    mysql> select ename, hiredate from emp where month(hiredate) = 12;

```
+-------+------------+
| ename | hiredate   |
```

```
+-------+------------+
| SMITH | 1980-12-17 |
| AARAV | 1981-12-31 |
| SCOTT | 1982-12-09 |
| JAMES | 1981-12-03 |
| FORD  | 1981-12-03 |
+-------+------------+
5 rows in set (0.05 sec)
```

19. *List names and hiredate of employees hired in the year 1980*

```
mysql> select ename, hiredate from emp where year(hiredate) = 1980;
+--------+------------+
| ename  | hiredate   |
+--------+------------+
| SMITH  | 1980-12-17 |
| GRASS  | 1980-02-14 |
| AARUSH | 1980-02-14 |
+--------+------------+
3 rows in set (0.03 sec)
```

20. *Display names and jobs of the people separated by a hyphen. Capitalize the first character of name and job.*

```
mysql> select concat(ucase(left(ename,1)), lcase(substr(ename,2)),
    ->          ' - ',
    ->          ucase(left(job,1)), lcase(substr(job,2))) as NameJob
    -> from emp;
+--------------------------------+
| NameJob                        |
+--------------------------------+
| Gita - Compliance officers     |
| Vrushali - Compliance officers |
| Sharmin - Public relation      |
| Martin - Salesman              |
| Blake - Manager                |
| Clark - Manager                |
| Miller - Clerk                 |
| Aarush - Salesman              |
| Supriya - Analyst              |
+--------------------------------+
30 rows in set (0.04 sec)
```

21. *List employee numbers, names and hiredates of the people working in the department number 20, display the hiredates in the dd/mm/yy format*

    mysql> select empno, ename, date_format(hiredate, '%d/%m/%y') as HireDate from emp where deptno = 20;

```
+-------+---------+----------+
| empno | ename   | HireDate |
+-------+---------+----------+
|  6473 | SHARMIN | 19/07/81 |
|  6781 | BANDISH | 19/07/81 |
|  7369 | SMITH   | 17/12/80 |
|  7566 | JONES   | 02/04/81 |
|  7788 | SCOTT   | 09/12/82 |
|  7840 | FRED    | 09/08/83 |
|  7876 | ADAMS   | 12/01/83 |
|  7902 | FORD    | 03/12/81 |
+-------+---------+----------+
8 rows in set (0.04 sec)
```

22. *Find number of months the president has worked for the company.*

    mysql> select ename, hiredate, timestampdiff(month, hiredate, curdate()) as MonthsWorked from emp where job = 'PRESIDENT';

```
+-------+------------+--------------+
| ename | hiredate   | MonthsWorked |
+-------+------------+--------------+
| KING  | 1981-11-17 |          533 |
+-------+------------+--------------+
1 row in set (0.06 sec)
```

23. *Find the day of the week on which SMITH joined*

    mysql> select ename, hiredate, dayname(hiredate) as DayOfWeek from emp where ename = 'SMITH';

```
+-------+------------+-----------+
| ename | hiredate   | DayOfWeek |
+-------+------------+-----------+
| SMITH | 1980-12-17 | Wednesday |
+-------+------------+-----------+
1 row in set (0.04 sec)
```

24. *Find the time of time of the day in which ADAMS joined*

    mysql> select ename, hiredate, time(hiredate) as TimeOfDay from emp where ename = 'ADAMS';

```
+-------+------------+-----------+
| ename | hiredate   | TimeOfDay |
```

```
+-------+------------+------------+
| ADAMS | 1983-01-12 | 00:00:00  |
+-------+------------+------------+
1 row in set (0.00 sec)
```

25. *Find day of month on which KING joined*

mysql> select ename, hiredate, day(hiredate) as DayOfMonth from emp where ename = 'KING';

```
+-------+------------+------------+
| ename | hiredate   | DayOfMonth |
+-------+------------+------------+
| KING  | 1981-11-17 |        17 |
+-------+------------+------------+
1 row in set (0.00 sec)
```

26. *Find out month on which MARTIN joined*

mysql> select ename, hiredate, monthname(hiredate) as MonthJoined from emp where ename = 'MARTIN';

```
+--------+------------+-------------+
| ename  | hiredate   | MonthJoined |
+--------+------------+-------------+
| MARTIN | 1981-09-28 | September   |
+--------+------------+-------------+
1 row in set (0.00 sec)
```

27. *Find out which quarter of the year the employees joined. Display their number and names as well*

mysql> select empno, ename, quarter(hiredate) as QuarterJoined from emp;

```
+-------+----------+---------------+
| empno | ename    | QuarterJoined |
+-------+----------+---------------+
|  6001 | GITA     |             3 |
|  6129 | VRUSHALI |             3 |
|  6473 | SHARMIN  |             3 |
|  6781 | BANDISH  |             3 |
|  7369 | SMITH    |             4 |
|  7415 | AARAV    |             4 |
|  7421 | THOMAS   |             3 |
|  9400 | SANGITA  |             3 |
|  9473 | SUPRIYA  |             3 |
+-------+----------+---------------+
30 rows in set (0.00 sec)
```

28. *Retrieve ANALYST records with the hiredate formatted as – 'The 3rd of December 1984'*

```
select ename,
    concat('The ', day(hiredate),
        case day(hiredate)
            when 1 then 'st'
            when 2 then 'nd'
            when 3 then 'rd'
            else 'th'
        end,
        ' of ', monthname(hiredate), ' ', year(hiredate)) as FormattedHireDate
from emp
where job = 'ANALYST';
+---------+-------------------------+
| ename   | FormattedHireDate       |
+---------+-------------------------+
| SCOTT   | The 9th of December 1982 |
| FORD    | The 3rd of December 1981 |
| SUPRIYA | The 19th of July 1981    |
+---------+-------------------------+
3 rows in set (0.04 sec)
```

29. *List all names, jobs, and a job classification number, which is to be assigned by you. Translate the value started in each job field to a job classification number.*
This is to be done as follows-
A. CLERK
B. MANAGER
C. PRESIDENT
D. OTHER

```
select ename, job,
    case job
        when 'CLERK' then 'A'
        when 'MANAGER' then 'B'
        when 'PRESIDENT' then 'C'
        else 'D'
    end as JobClass
from emp;
+----------+--------------------+----------+
| ename    | job                | JobClass |
+----------+--------------------+----------+
| GITA     | Compliance officers | D        |
```

```
| VRUSHALI | Compliance officers | D       |
| SHARMIN  | Public Relation     | D       |
| SUPRIYA  | ANALYST             | D       |
+----------+---------------------+----------+
30 rows in set (0.00 sec)
```

30. *Display the length of the longest employees name*

```
mysql> select max(length(ename)) as LongestNameLength from emp;
+-------------------+
| LongestNameLength |
+-------------------+
|                 8 |
+-------------------+
1 row in set (0.00 sec)
```

31. *Write a query to list the length of service of the employees (of the form n years and m months).*

```
select ename,
    concat(
      floor(timestampdiff(month, hiredate, curdate())/12), ' years and ',
      mod(timestampdiff(month, hiredate, curdate()), 12), ' months'
    ) as LengthOfService
from emp;
+----------+------------------------+
| ename    | LengthOfService        |
+----------+------------------------+
| GITA     | 44 years and 9 months  |
| VRUSHALI | 44 years and 9 months  |
| SHARMIN  | 44 years and 9 months  |
| BANDISH  | 44 years and 9 months  |
| SANGITA  | 44 years and 9 months  |
| SUPRIYA  | 44 years and 9 months  |
+----------+------------------------+
30 rows in set (0.00 sec)
```

32. *How many employees who are joined in 1985.*

```
mysql> select count(*) as EmployeesJoined1985
 -> from emp
-> where year(hiredate) = 1985;
+---------------------+
| EmployeesJoined1985 |
+---------------------+
|                   0 |
+---------------------+
1 row in set (0.00 sec)
```

33. *How many employees joined each month in 1985.*

```
select monthname(hiredate) as MonthJoined,
    count(*) as EmployeesCount
from emp
where year(hiredate) = 1985
group by month(hiredate), monthname(hiredate)
order by month(hiredate);
Empty set (0.00 sec)
```

34. *How many employees who are joined in March 1985.*

```
select count(*) as EmployeesJoinedMarch1985
from emp
where year(hiredate) = 1985
  and month(hiredate) = 3;
+--------------------------+
| EmployeesJoinedMarch1985 |
+--------------------------+
|                        0 |
+--------------------------+
1 row in set (0.00 sec)
```

35. *Find the total sales amount*

```
mysql> select sum(sal + ifnull(comm,0)) as TotalSalesAmount from emp where job =
'SALESMAN';
+------------------+
| TotalSalesAmount |
+------------------+
|            17125 |
+------------------+
1 row in set (0.00 sec)
```

36. *Find the customer-wise lowest and highest sales amount*

```
select custno,
    min(salesamt) as LowestSales,
    max(salesamt) as HighestSales
from sales group by custno
order by custno;
+------------------+
| TotalSalesAmount |
+------------------+
|            17125 |
+------------------+
1 row in set (0.00 sec)
```

37. *Find product-wise lowest, highest and total sales.*

```
select prodid, prodname,
    min(amount) as LowestSales,
    max(amount) as HighestSales,
    sum(amount) as TotalSales
from sales
group by prodid, prodname
order by prodid;
+--------+-------------------------+-------------+--------------+------------+
| prodid | prodname                | LowestSales | HighestSales | TotalSales |
+--------+-------------------------+-------------+--------------+------------+
| 100860 | ACE TENNIS RACKET I     |       35.00 |     16569.00 |   22144.00 |
| 100861 | ACE TENNIS RACKET II    |       84.00 |      4500.00 |    8809.50 |
| 100870 | ACE TENNIS BALLS-3 PACK |        8.40 |      3306.00 |    5426.90 |
| 100871 | ACE TENNIS BALLS-6 PACK |      250.00 |      5600.00 |   10056.80 |
| 100890 | ACE TENNIS NET          |       50.00 |     29000.00 |   29398.00 |
| 101860 | SP TENNIS RACKET        |       24.00 |      4800.00 |    7224.00 |
| 101863 | SP JUNIOR RACKET        |     1500.00 |      2500.00 |    4000.00 |
| 102130 | RH: "GUIDE TO TENNIS"   |       34.00 |      1703.40 |    2417.40 |
| 200376 | SB ENERGY BAR-6 PACK    |        2.40 |      2400.00 |    3586.40 |
| 200380 | SB VITA SNACK-6 PACK    |      400.00 |      1200.00 |    2200.00 |
+--------+-------------------------+-------------+--------------+------------+
10 rows in set (0.04 sec)
```

38. *Find department-wise average salary for all the departments employing more than three employees*

```
mysql> select deptno,  avg(sal) as AvgSalary from emp group by deptno having count(*) > 3
order by deptno;
+--------+-----------+
| deptno | AvgSalary |
+--------+-----------+
|     10 | 2770.0000 |
|     20 | 2421.8750 |
|     30 | 1852.0833 |
+--------+-----------+
3 rows in set (0.00 sec)
```

39. *Find the customer-wise total sales for all the customers except 'TKB SPORT SHOP' who came to purchase various sports items maximum four times.*

```
select custid, custname,
    sum(amount) as TotalSales,
    count(*) as PurchaseCount
from sales
where custname <> 'TKB SPORT SHOP'
```

```
group by custid, custname
having count(*) <= 4
order by custid;
+--------+---------------+------------+---------------+
| custid | custname      | TotalSales | PurchaseCount |
+--------+---------------+------------+---------------+
|    102 | VOLLYRITE     |   27775.50 |             4 |
|    106 | SHAPE UP      |     700.40 |             4 |
|    107 | WOMENS SPORTS |     710.00 |             3 |
|    108 | NORTH WOODS   |    6400.00 |             4 |
+--------+---------------+------------+---------------+
4 rows in set (0.00 sec)
```

40. *Display the highest, lowest, sum and average salary for all employees. Label the columns appropriately.*

```
select
    max(sal) as HighestSalary,
    min(sal) as LowestSalary,
    sum(sal) as TotalSalary,
    avg(sal) as AverageSalary
from emp;
+---------------+--------------+-------------+---------------+
| HighestSalary | LowestSalary | TotalSalary | AverageSalary |
+---------------+--------------+-------------+---------------+
|          5000 |          800 |       63700 |     2123.3333 |
+---------------+--------------+-------------+---------------+
1 row in set (0.00 sec)
```

41. *Modify the above query and display the output for each job type.*

```
mysql> select job,
    max(sal) as HighestSalary,
    min(sal) as LowestSalary,
    sum(sal) as TotalSalary,
    avg(sal) as AverageSalary
from emp
group by job
order by job;
+---------------------+---------------+--------------+-------------+---------------+
| job                 | HighestSalary | LowestSalary | TotalSalary | AverageSalary |
+---------------------+---------------+--------------+-------------+---------------+
| ANALYST             |          3000 |         1750 |        7750 |     2583.3333 |
| vice PRESIDENT      |          5000 |         1000 |        6000 |     3000.0000 |
+---------------------+---------------+--------------+-------------+---------------+
9 rows in set (0.00 sec)
```

42. *List names of people who have salary less than the average salary for dept 20*

```
select ename, sal
from emp
where deptno = 20
  and sal < (
     select avg(sal)
     from emp
     where deptno = 20
  );
+---------+------+
| ename   | sal  |
+---------+------+
| SHARMIN | 1750 |
| BANDISH | 1750 |
| SMITH   |  800 |
| ADAMS   | 1100 |
+---------+------+
4 rows in set (0.00 sec)
```

43. *Find the average annual salary per job in each department.*

```
select deptno,  job,  avg(sal) * 12 as AvgAnnualSalary from emp group by deptno, job order by deptno, job;
+--------+--------------------+-----------------+
| deptno | job                | AvgAnnualSalary |
+--------+--------------------+-----------------+
|    10 | CLERK              |     25600.0000 |
|    10 | MANAGER            |     29400.0000 |
|    10 | PRESIDENT          |     60000.0000 |
|    20 | ANALYST            |     36000.0000 |
|    60 | vice PRESIDENT     |     12000.0000 |
+--------+--------------------+-----------------+
16 rows in set (0.00 sec)
```

44. *Count the number of people in department 30 who receive a salary and the number of people who receive a commission*

```
mysql> select count(sal) as SalaryCount,  count(comm) as CommissionCount from emp where deptno = 30;
+-------------+----------------------------+
| SalaryCount | CommissionCount |
+-------------+---------------------------+
|        12 |            7                        |
+-------------+---------------------------+
1 row in set (0.00 sec)
```

45. *Compute the average, minimum and maximum salaries of these groups of employees having job as Clerk or manager, Display the job as well*

mysql> select job, avg(sal) as AverageSalary, min(sal) as MinimumSalary, max(sal) as MaximumSalary from emp where job in ('CLERK', 'MANAGER') group by job order by job;

```
+---------+---------------+---------------+---------------+
| job     | AverageSalary | MinimumSalary | MaximumSalary |
+---------+---------------+---------------+---------------+
| CLERK   |    1571.4286 |          800 |          3350 |
| MANAGER |    2735.0000 |         1250 |          4150 |
+---------+---------------+---------------+---------------+
2 rows in set (0.00 sec)
```

46. *Write an SQL command that displays 2nd highest salary paid*

mysql> select max(sal) as SecondHighestSalary from emp where sal < (select max(sal) from emp);

```
+---------------------+
| SecondHighestSalary |
+---------------------+
|              4150 |
+---------------------+
1 row in set (0.00 sec)
```

47. *Write a query to find the employees who are earning the maximum salary in their departments.*

mysql> select ename, deptno, sal from emp e where sal = (select max(sal) from emp where deptno = e.deptno);

```
+----------+--------+------+
| ename    | deptno | sal  |
+----------+--------+------+
| VRUSHALI |     50 | 1750 |
| KING     |     10 | 5000 |
| FRED     |     20 | 5000 |
| Ruhan    |     60 | 2000 |
| HOFFMAN  |     30 | 4150 |
| SANGITA  |     50 | 1750 |
| SUPRIYA  |     50 | 1750 |
+----------+--------+------+
7 rows in set (0.01 sec)
```

48. *Write a query to find the salesman number (repid) who has achieved the maximum total sales among the entire salesman.*

mysql> select repid from sales group by repid order by sum(amount) desc limit 1;

```
+-------+
| repid |
+-------+
|  7844 |
+-------+
1 row in set (0.00 sec)
```

49. *List the highest salary paid for each job.*

mysql> select job, max(sal) as HighestSalary from emp group by job order by job;

```
+---------------------+---------------+
| job                 | HighestSalary |
+---------------------+---------------+
| ANALYST             |          3000 |
| CLERK               |          3350 |
| SALESMAN            |          2575 |
| vice PRESIDENT      |          5000 |
+---------------------+---------------+
9 rows in set (0.00 sec)
```

50. *Find the most recently hired employee in each department.*

mysql> select ename, deptno, hiredate from emp e where hiredate = (select max(hiredate) from emp where deptno = e.deptno);

```
+----------+--------+------------+
| ename    | deptno | hiredate   |
+----------+--------+------------+
| VRUSHALI |     50 | 1981-07-19 |
| FRED     |     20 | 1983-08-09 |
| MILLER   |     10 | 1982-01-23 |
| SANGITA  |     50 | 1981-07-19 |
| SUPRIYA  |     50 | 1981-07-19 |
+----------+--------+------------+
8 rows in set (0.00 sec)
```

51. *In which year did most people join the company? Display the year and the number of employees.*

mysql> select year(hiredate) as JoinYear, count(*) as NumEmployees from emp group by year(hiredate) order by NumEmployees desc limit 1;

```
+----------+--------------+
| JoinYear | NumEmployees |
+----------+--------------+
|     1981 |           19 |
+----------+--------------+
1 row in set (0.00 sec)
```

52. *Write a query to display employee name whose name occurs only once in the table.*

mysql> select ename from emp group by ename having count(*) = 1;

```
+----------+
| ename    |
+----------+
| GITA     |
| VRUSHALI |
| SCOTT    |
```

```
| SUPRIYA  |
+----------+
30 rows in set (0.00 sec)
```

53. *Write a query to display all the details from dept table along with the no. of employee working in each dept.*

```
mysql> select d.deptno,
    ->      d.dname,
    ->      d.loc,
    ->      count(e.empno) as NumEmployees
    -> from dept d
    -> left join emp e
    ->      on d.deptno = e.deptno
    -> group by d.deptno, d.dname, d.loc;
+--------+------------+----------+--------------+
| deptno | dname      | loc      | NumEmployees |
+--------+------------+----------+--------------+
|     10 | ACCOUNTING | NEW YORK |            5 |
|     20 | RESEARCH   | DALLAS   |            8 |
|     30 | SALES      | CHICAGO  |           12 |
|     40 | OPERATIONS | BOSTON   |            0 |
|     50 | HR         | PUNE     |            3 |
|     60 | IT         | BARODA   |            2 |
+--------+------------+----------+--------------+
```

54. *Find out which department does not have any employees.*

```
mysql> select d.deptno, d.dname, d.loc from dept d
    -> left join emp e on d.deptno = e.deptno
    -> where e.empno is null;
+--------+------------+--------+
| deptno | dname      | loc    |
+--------+------------+--------+
|     40 | OPERATIONS | BOSTON |
+--------+------------+--------+
1 row in set (0.00 sec)
```

55. *List out the no. of employees joined in every month in ascending order.*

```
mysql> select month(hiredate) as JoinMonth,
    ->      count(*) as NumEmployees
    -> from emp
    -> group by month(hiredate)
    -> order by JoinMonth;
+-----------+--------------+
| JoinMonth | NumEmployees |
```

```
+-----------+--------------+
|       1 |          3 |
|       2 |          4 |
|       3 |          1 |
|       4 |          1 |
|       5 |          1 |
|       6 |          1 |
|       7 |          8 |
|       8 |          3 |
|       9 |          2 |
|      11 |          1 |
|      12 |          5 |
+-----------+--------------+
11 rows in set (0.00 sec)
```